

Locally periodic Timoshenko rod: Experiment and theory

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The flexural vibrations of a locally periodic rod, which consists of N unit cells, are discussed both from the experimental and theoretical points of view. Timoshenko's beam theory and the transfer matrix method are used to calculate the normal-mode frequencies and amplitudes. The theoretical values are then compared with the experimental ones, which are obtained using an electromagnetic acoustic transducer (EMAT). Good agreement between the numerical results and the experimental measurements is obtained. It is shown that as N grows, a band spectrum emerges. © 2005 Acoustical Society of America. [DOI: 10.1121/1.1880732]

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I. INTRODUCTION

It is well known that spatially periodic systems show a spectrum which is arranged in bands of allowed frequencies, or pass bands, separated by gaps of forbidden frequencies, called stop bands.¹ One-dimensional locally periodic systems, composed of a finite number N of identical cells, also show a spectrum in which the normal-mode frequencies are clustered in bands as N grows. This has been obtained theoretically for a variety of wave-like systems: electrons in solids,² elastic systems,^{3,4} Love analysis of waves in rods,⁵ waves in a stretched wire,⁶ and compressional and torsional waves in rods.⁷ It should be noted that, except for the last two references, a direct comparison with experimental data is not performed. Apart from its academic value, some of these results have applications in vibration isolation.⁸

The emergence of a banded spectrum is obtained in all these cases for systems related to second-order differential equations. Flexural vibrations, described by a more complicated fourth-order differential equation,⁹ have also been analyzed from the theoretical standpoint for an infinite supported Timoshenko beam^{10,11} and numerically for a multi-span beam using the finite element method.¹² However, in none of these works is a comparison with experiment performed. Therefore, our main goal in this paper is to study flexural vibrations in locally periodic rods from the theoretical and experimental viewpoints.

The Timoshenko beam theory (TBT) as well as the simpler Bernoulli-Euler (B-E) approach will be analyzed. In Sec. II the transfer matrix method is used to calculate both normal-mode frequencies and amplitudes for these rods. In Sec. III the experimental setup to measure these quantities is described. An electromagnetic acoustic transducer (EMAT) is used to excite and detect the rod vibrations.¹³ This EMAT is inexpensive and easy to construct with commercially available components. It is a non-contact and very versatile device which operates at low frequencies. Furthermore, this

EMAT is selective in the sense that either compressional, torsional or flexural vibrations can be excited or detected with it. Finally, in Sec. IV the theoretical results are compared with the experimental ones. Since a good agreement is obtained, this result works as another check of TBT, which was formerly accomplished only for a uniform rod.^{14–16}

II. TRANSFER MATRIX METHOD FOR THE TIMOSHENKO EQUATION

There are two well-known theories which describe flexural vibrations of beams, the Bernoulli-Euler and the Timoshenko beam theories.⁹ In the TBT, apart from the transverse displacement ξ , another degree of freedom exists, the angular coordinate ψ .⁹ The Timoshenko equation for the transverse displacement ξ of a uniform rod is given by

$$\frac{EI}{\rho S} \frac{\partial^4 \xi}{\partial z^4} - \frac{I}{S} \left(1 + \frac{E}{\kappa G} \right) \frac{\partial^4 \xi}{\partial z^2 \partial t^2} + \frac{\partial^2 \xi}{\partial t^2} + \frac{\rho I}{\kappa GS} \frac{\partial^4 \xi}{\partial t^4} = 0. \quad (1)$$

An identical equation holds for the angle ψ . Here G and E are the shear and Young's modulus, respectively, ρ is the density, S is the cross-section area, κ is the Timoshenko shear coefficient and I is the moment of inertia with respect to the axis perpendicular to the rod axis. Imposing the normal-mode condition $\xi(z, t) = \Psi(z) \exp(i\omega t)$, where $\omega = 2\pi f$, f being the frequency, the Timoshenko equation can be written as a set of four linear differential equations.¹⁶ As has been shown in Ref. 16, for a circular uniform rod the amplitude $\Psi(z)$ reads as

$$\Psi(z) = Ak_1^{-3} e^{k_1 z} + Bk_2^{-3} e^{k_2 z} + Ck_3^{-3} e^{k_3 z} + Dk_4^{-3} e^{k_4 z}, \quad (2)$$

where $z=0$ and $z=L$ correspond, respectively, to the left- and right-hand free ends of the rod. In the last equation,

$$k_j = (-1)^{[j/2]} \sqrt{\frac{\alpha + (-1)^j \sqrt{\alpha^2 + 4\beta}}{2}}, \quad \text{for } j = 1, \dots, 4, \quad (3)$$

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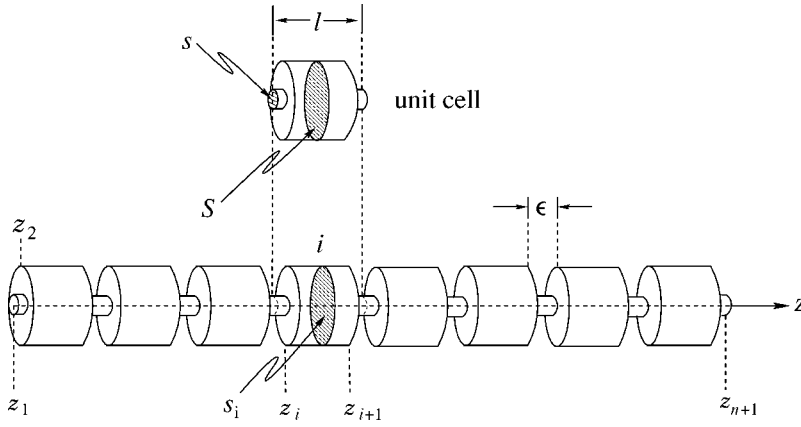


FIG. 1. A schematic diagram of the locally periodic rod formed by N unit cells of length l , with $n = 2N + 1$ cylinders. The total length L is Nl .

depends on the mechanical and geometrical properties of the rod through $\alpha = -\rho\omega^2/E - \rho\omega^2/\kappa G$ and $\beta = \rho S\omega^2/EI - \rho^2\omega^4/E\kappa G$. Here $[j/2]$ is the largest integer less than $j/2$.

The eigenfrequencies are obtained when the free-end boundary conditions are fulfilled. Then the bending moment,

$$M(z) = EI\Psi''(z), \quad (4)$$

and the shear force,

$$F(z) = EI\Psi'''(z), \quad (5)$$

vanish at $z=0$ and $z=L$. This leads to

$$\begin{aligned} k_1^{-1}A + k_2^{-1}B + k_3^{-1}C + k_4^{-1}D &= 0, \\ A + B + C + D &= 0, \\ k_1^{-1}e^{k_1L}A + k_2^{-1}e^{k_2L}B + k_3^{-1}e^{k_3L}C + k_4^{-1}e^{k_4L}D &= 0, \\ e^{k_1L}A + e^{k_2L}B + e^{k_3L}C + e^{k_4L}D &= 0. \end{aligned} \quad (6)$$

The eigenfrequencies f_i are then found obtaining the roots of the determinant of this system numerically.

These results are now used to deal with the locally periodic rod. The system is formed by coaxial cylinders of areas S and s with $S > s$, as shown in Fig. 1. As was done in Ref. 7, the transfer matrix method is used. There compressional and torsional waves were studied and the transfer matrix was 2×2 , because these oscillations obey a second-order differential equation. Since flexural vibrations are described by a fourth-order differential equation, the transfer matrix is now 4×4 .

If the notation indicated in Fig. 1 is used, the amplitude $\Psi_i(z)$ in the i th cylinder is

$$\begin{aligned} \Psi_i(z) &= A_i k_{1,i}^{-3} e^{k_{1,i}(z-z_i)} + B_i k_{2,i}^{-3} e^{k_{2,i}(z-z_i)} \\ &+ C_i k_{3,i}^{-3} e^{k_{3,i}(z-z_i)} + D_i k_{4,i}^{-3} e^{k_{4,i}(z-z_i)}, \end{aligned} \quad (7)$$

which implies the use of a local coordinate system. Here $z_i \leq z \leq z_{i+1}$, $i = 1, \dots, n$ and $k_{j,i}$ with $j = 1, \dots, 4$ is now given by

$$k_{j,i} = (-1)^{[j/2]} \sqrt{\frac{\alpha_i + (-1)^j \sqrt{\alpha_i^2 + 4\beta_i}}{2}}, \quad (8)$$

where

$$\alpha_i = -\rho_i\omega^2/E_i - \rho_i\omega^2/\kappa_i G_i \quad (9)$$

and

$$\beta_i = \rho_i S_i \omega^2/E_i I_i - \rho_i^2 \omega^4/E_i \kappa_i G_i. \quad (10)$$

The transfer matrix for the B-E case is obtained replacing the right-hand side of Eq. (8) by $(-1)^{[j/2]} \sqrt{(-1)^j \omega \sqrt{S_i/E_i I_i}}$. Note that $k_{j,i}$ depends on the radius of each cylinder so two characteristic wavelengths exist for the rod of Fig. 1.

The conditions of continuity at the points where the cylinders of different radii join each other are as follows: continuity of the displacement, slope, bending moment Eq. (4) and shear force Eq. (5):

$$\begin{aligned} \Psi_i|_{z_{i+1}} &= \Psi_{i+1}|_{z_{i+1}}, \\ \Psi'_i|_{z_{i+1}} &= \Psi'_{i+1}|_{z_{i+1}}, \end{aligned} \quad (11)$$

$$E_i I_i \Psi''_i|_{z_{i+1}} = E_{i+1} I_{i+1} \Psi''_{i+1}|_{z_{i+1}},$$

$$E_i I_i \Psi'''_i|_{z_{i+1}} = E_{i+1} I_{i+1} \Psi'''_{i+1}|_{z_{i+1}}.$$

The amplitude $\Psi_i(z)$ and its derivatives depend on the physical and geometrical properties of cylinder i through Eqs. (8) to (10). It should be pointed that these continuity conditions are approximate.^{7,17,18}

Using Eq. (7) in the continuity conditions, the coefficients A_{i+1} , B_{i+1} , C_{i+1} , and D_{i+1} in the $(i+1)$ th cylinder can be expressed in terms of A_i , B_i , C_i , and D_i in the following form:

$$(A_{i+1}, B_{i+1}, C_{i+1}, D_{i+1})^T = \mathbf{M}_{i \rightarrow i+1} (A_i, B_i, C_i, D_i)^T, \quad (12)$$

where T means taking the transpose and

$$\mathbf{M}_{i \rightarrow i+1} = \begin{pmatrix} k_{1,i+1}^{-3} & k_{2,i+1}^{-3} & k_{3,i+1}^{-3} & k_{4,i+1}^{-3} \\ k_{1,i+1}^{-2} & k_{2,i+1}^{-2} & k_{3,i+1}^{-2} & k_{4,i+1}^{-2} \\ k_{1,i+1}^{-1} & k_{2,i+1}^{-1} & k_{3,i+1}^{-1} & k_{4,i+1}^{-1} \\ 1 & 1 & 1 & 1 \end{pmatrix}^{-1} \times \begin{pmatrix} k_{1,i}^{-3} e^{k_{1,i}(z_{i+1}-z_i)} & k_{2,i}^{-3} e^{k_{2,i}(z_{i+1}-z_i)} & k_{3,i}^{-3} e^{k_{3,i}(z_{i+1}-z_i)} & k_{4,i}^{-3} e^{k_{4,i}(z_{i+1}-z_i)} \\ k_{1,i}^{-2} e^{k_{1,i}(z_{i+1}-z_i)} & k_{2,i}^{-2} e^{k_{2,i}(z_{i+1}-z_i)} & k_{3,i}^{-2} e^{k_{3,i}(z_{i+1}-z_i)} & k_{4,i}^{-2} e^{k_{4,i}(z_{i+1}-z_i)} \\ \eta_i k_{1,i}^{-1} e^{k_{1,i}(z_{i+1}-z_i)} & \eta_i k_{2,i}^{-1} e^{k_{2,i}(z_{i+1}-z_i)} & \eta_i k_{3,i}^{-1} e^{k_{3,i}(z_{i+1}-z_i)} & \eta_i k_{4,i}^{-1} e^{k_{4,i}(z_{i+1}-z_i)} \\ \eta_i e^{k_{1,i}(z_{i+1}-z_i)} & \eta_i e^{k_{2,i}(z_{i+1}-z_i)} & \eta_i e^{k_{3,i}(z_{i+1}-z_i)} & \eta_i e^{k_{4,i}(z_{i+1}-z_i)} \end{pmatrix}, \quad (13)$$

with $\eta_i = I_i / I_{i+1}$. Applying successively $\mathbf{M}_{1 \rightarrow 2}$, $\mathbf{M}_{2 \rightarrow 3}$, until $\mathbf{M}_{(n-1) \rightarrow n}$, it follows that

$$(A_n, B_n, C_n, D_n)^T = \mathbf{M}(A_1, B_1, C_1, D_1)^T, \quad (14)$$

where the transfer matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \mathbf{M}_{(n-1) \rightarrow n} \mathbf{M}_{(n-2) \rightarrow (n-1)} \cdots \mathbf{M}_{2 \rightarrow 3} \mathbf{M}_{1 \rightarrow 2}. \quad (15)$$

The appropriate boundary conditions for free ends at $z_1 = 0$ and $z_{n+1} = L$ are given by the vanishing of the bending moment and the shear force. These conditions imply the following algebraic equations:

$$\begin{aligned} k_{1,1}^{-1} A_1 + k_{2,1}^{-1} B_1 + k_{3,1}^{-1} C_1 + k_{4,1}^{-1} D_1 &= 0, \\ A_1 + B_1 + C_1 + D_1 &= 0, \\ k_{1,n}^{-1} e^{\mu_{1,n}} A_n + k_{2,n}^{-1} e^{\mu_{2,n}} B_n + k_{3,n}^{-1} e^{\mu_{3,n}} C_n + k_{4,n}^{-1} e^{\mu_{4,n}} D_n &= 0, \\ e^{\mu_{1,n}} A_n + e^{\mu_{2,n}} B_n + e^{\mu_{3,n}} C_n + e^{\mu_{4,n}} D_n &= 0, \end{aligned} \quad (16)$$

where $\mu_{j,n} = k_{j,n}(z_{n+1} - z_n)$, with $j = 1, \dots, 4$. If the coefficients A_n , B_n , C_n and D_n of Eq. (14) are substituted in Eqs. (16) the system of four homogeneous equations can be written as

$$\mathbf{P}(A_1, B_1, C_1, D_1)^T = 0, \quad (17)$$

where the matrix $\mathbf{P}$ is given by

$$\mathbf{P} = \begin{pmatrix} k_{1,1}^{-1} & k_{2,1}^{-1} & k_{3,1}^{-1} & k_{4,1}^{-1} \\ 1 & 1 & 1 & 1 \\ \sum_{i=1}^4 t_{i,1} & \sum_{i=1}^4 t_{i,2} & \sum_{i=1}^4 t_{i,3} & \sum_{i=1}^4 t_{i,4} \\ \sum_{i=1}^4 h_{i,1} & \sum_{i=1}^4 h_{i,2} & \sum_{i=1}^4 h_{i,3} & \sum_{i=1}^4 h_{i,4} \end{pmatrix}, \quad (18)$$

with $t_{i,j} = M_{ij} k_{i,n}^{-1} e^{k_{i,n}(z_{n+1}-z_n)}$ and $h_{i,j} = M_{ij} e^{k_{i,n}(z_{n+1}-z_n)}$. Here M_{ij} are the matrix elements of $\mathbf{M}$. The normal mode frequencies are then finally obtained setting $\det(\mathbf{P}) = 0$.

III. THE EXPERIMENT

A block diagram of the experimental setup is shown in Fig. 2. The signal of an oscillator is amplified and sent to the EMAT exciter. The detector signal is converted by a lock-in amplifier into a dc voltage proportional to the amplitude. The detector is moved along the rod to obtain the amplitude as a

function of z . A somewhat more detailed description of this fully automatized system is given in Ref. 16.

The EMAT used to detect or excite waves in rods consists of a coil and a permanent magnet. It employs Foucault (or eddy) currents, as indicated in Refs. 7, 13, 16. The EMAT is shown in the left lower corner of Fig. 2, where it is configured to detect the amplitude ξ of the flexural wave, without any mechanical contact with the rod.

The results obtained by measuring ξ can be verified with a different configuration of the EMAT, shown in Fig. 3. With this configuration it is possible to detect the angular coordinate ψ of the flexural mode. The two permanent magnets produce a magnetic field parallel to the X axis. Due to this field, any rotation of the rod with respect to the Y axis induces a Foucault current in the XY plane. This in turn generates a magnetic field along the Z axis. The emf induced by this field in the coil is the signal corresponding to the angular mode. It is not difficult to understand that the same configuration can be used to excite the ψ oscillation. An ac current in the coil produces a magnetic field that induces Foucault currents circulating in the XY plane. These currents interact with the magnetic field of the permanent magnets, giving rise to a torque in the direction of the Y axis. This seems to be a unique experimental setup, in the sense that it makes it possible to excite or detect the angular mode ψ along the rod

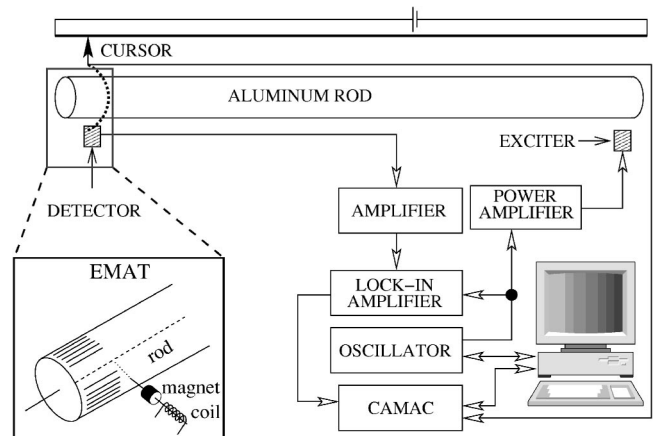


FIG. 2. Block diagram of the experimental setup. The EMATs (exciter and scanning detector) are shown shaded and in a configuration to excite or detect flexural modes. The dotted line connecting the cursor with the detector indicates mechanical coupling.

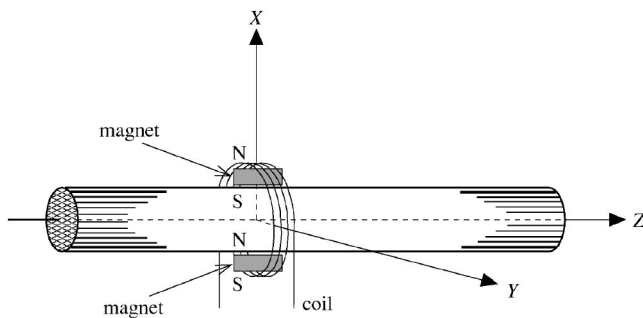


FIG. 3. The configuration of the coil and magnets of the EMAT to detect or excite the angular coordinate ψ .

without any mechanical contact between the rod and the exciter or detector.

The eigenfrequencies of a uniform aluminum rod of circular cross section have been measured with this setup. For each normal mode two very close resonances (doublets) are observed, as shown in Fig. 4. This is due to the lack of perfect circularity in the manufacturing process of the aluminum rods. The frequency difference between the two resonances of the doublet is very small, of the order of 0.2% of the frequency. This difference in frequency would correspond to a uniform rod of elliptical cross section with a difference between major and minor semi-axes of the order of $10\ \mu\text{m}$. This figure also shows the precision of our measurements. In what follows only one of the resonant frequencies will be given due to the scale used.

The measured eigenfrequencies are shown as open circles in Fig. 5 as a function of the mode number. They are compared with the TBT (triangles) and B-E (squares) predictions in the same figure. It is clearly seen that the simplest theory fails except at very low frequencies, as is well known.⁹ On the other hand, the TBT results agree very well with the experimental data.

IV. COMPARISON BETWEEN THEORY AND EXPERIMENT

Normal-mode amplitudes and frequencies were measured for locally periodic aluminum rods using the experi-

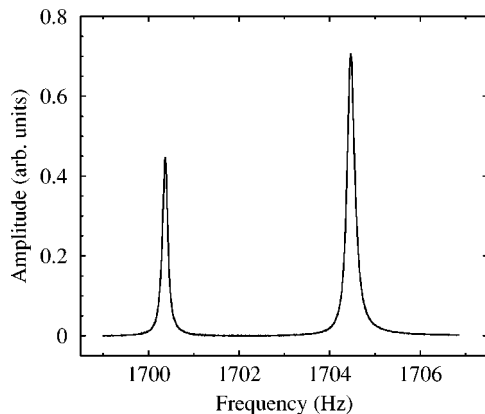


FIG. 4. Typical bending doublet measured for a uniform aluminum rod of circular cross section of radius 6.35 mm and length $L=0.9105\ \text{m}$. The frequency difference is small of the order of 0.2% of the normal-mode frequency f .

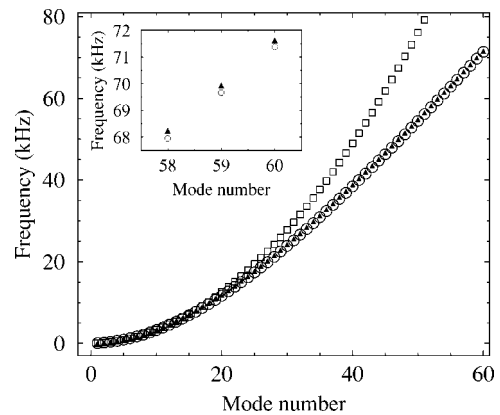


FIG. 5. A comparison between the measured normal-mode frequencies (open circles) with the Timoshenko beam theory results for $\kappa=0.925$ (triangles). The results obtained with the Bernoulli-Euler approach are given by the squares. The inset shows a zoom of some data. The same rod of Fig. 4 was used.

mental setup of Sec. III and the EMATs, as indicated in Figs. 2 and 3. The frequencies for $N=1, 4, 8$ and 12 unit cells are given in Fig. 6, where they are also compared to numerical values obtained both for the Timoshenko and Bernoulli-Euler theories. For each value of N the left column corresponds to the experimental values, the center column to the Timoshenko approach and the right-hand one to the B-E approximation. The numerical values were obtained using the transfer matrix method as explained in Sec. II. Except for the shear coefficient κ , the parameters entering the calculation were measured for each rod. The value $\kappa=(6+12\nu+6\nu^2)/(7+12\nu+4\nu^2)=0.925$ with the Poisson ratio $\nu=0.3$, a well known value for aluminum, was used.¹⁹ One could say, then, that these are parameter free calculations.

It is apparent from Fig. 6 that as N grows the normal-mode frequencies are grouped forming bands separated by gaps of forbidden frequencies. This is true for experimental values as well as for both theoretical approaches. As will be seen in this figure the Timoshenko predictions fit the measured values rather well, much better indeed that the Bernoulli-Euler results. The B-E theory agrees with experi-

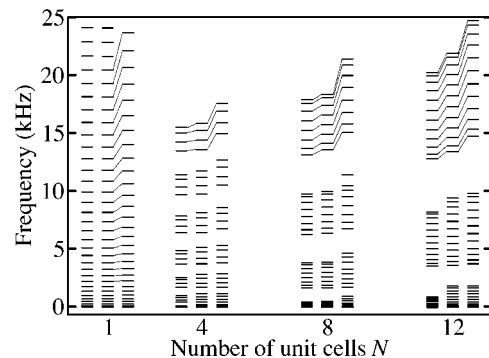


FIG. 6. Normal-mode frequencies for locally periodic rods with N unit cells. The left, center and right columns for each N correspond to the experimental values, Timoshenko and Bernoulli-Euler theoretical results, respectively. Four aluminum rods of length $L=1.005\ \text{m}$ and radius 6.35 mm were used. The notches were obtained machining the uniform rods and have radius $r=1.5\ \text{mm}$ and width $\epsilon=0.3\ \text{mm}$.

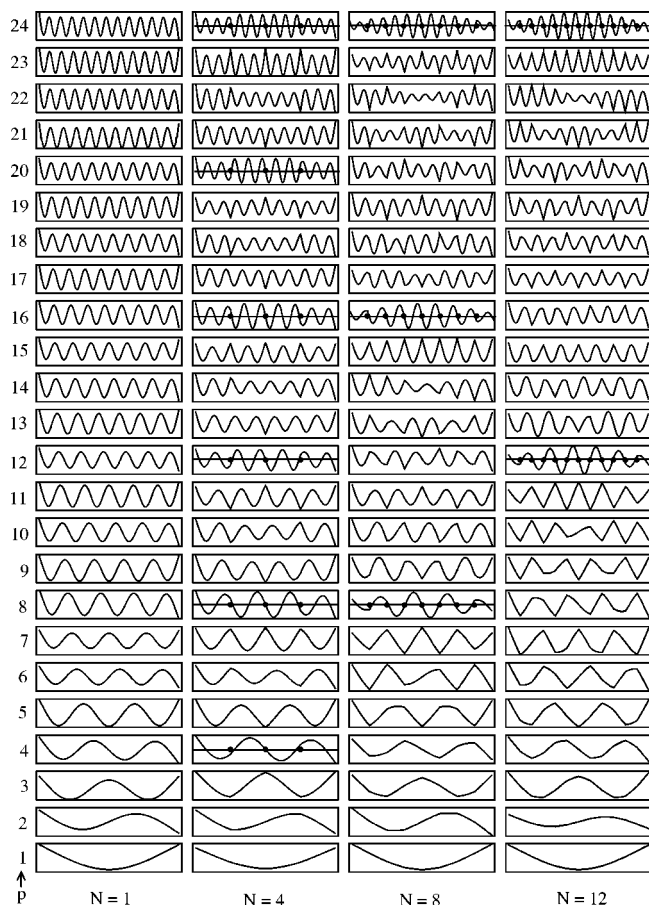


FIG. 7. Theoretical Timoshenko amplitudes for rods with $N-1$ notches. The amplitude vanishes at the position of the notches (dots) when the Bragg condition $p=mN$, m an integer, is fulfilled. The parameters of the rods are those of Fig. 6.

ment only at low frequencies, as could be expected from the results shown in Fig. 5 for $N=1$.

In Fig. 7, the theoretical amplitudes for the TBT are shown as a function of N for modes $p=1$ to $p=24$. The gap in the spectrum is produced when $p=mN$ with m an integer, i.e., when the Bragg condition $kl=m\pi$ is fulfilled. In this case, the amplitude vanishes at the location of the notches. The notches would then have a minimal effect and the corresponding state would have a normal-mode frequency very

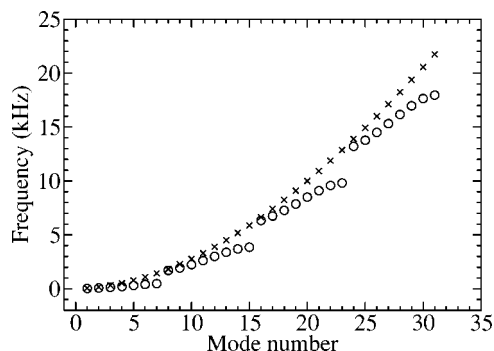


FIG. 8. A comparison of the flexural experimental frequencies of a uniform rod (crosses) with those of a rod with $N=8$ unit cells (open circles). The frequency shift is minimum when the Bragg condition is fulfilled. The parameters of the rods are the same as in Fig. 6.

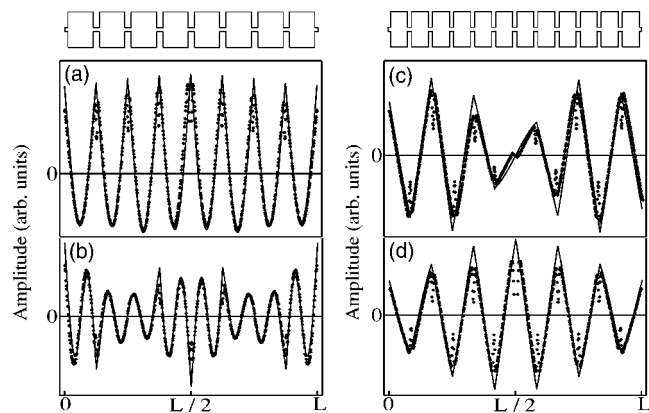


FIG. 9. Flexural amplitudes for locally periodic rods with $N=8$, (a) $p=15$, (b) $p=21$ and $N=12$, (c) $p=10$ and (d) $p=11$. Here p is the mode number. The continuous lines correspond to Timoshenko's theory and the dots to experimental measurements. The same parameters as in Fig. 6 are used. A sketch of the rod is shown in the upper part (not in scale) as a guide to the reader.

similar to the one corresponding to the $N=1$ case. This is shown to be the case in Fig. 8, where, as an example, the experimental values for $N=8$ are compared to the measurements for $N=1$. It is clear that the first mode of each band remains almost unaffected by the presence of the notches.

As mentioned before, the experimental setup makes it possible to measure the normal-mode amplitudes by moving the EMAT detector along the rod. A comparison between theoretical and experimental amplitudes is then achievable. A selection of these amplitudes are shown for $N=8$ and $N=12$ in Fig. 9. The theoretical predictions using TBT are shown as continuous lines and the measurements correspond to the dots. Except at the location of the notches, where the eddy currents detected by the EMAT are interrupted, the agreement between theory and experiment is excellent. It should be pointed out that the amplitudes arising from the B-E theory would give similar results since, as shown in Fig. 10, both theories predict very similar amplitudes, albeit different frequencies, as mentioned above.

V. CONCLUSIONS

Amplitudes for flexural vibrations in locally periodic rods were measured continuously along the rod for the first

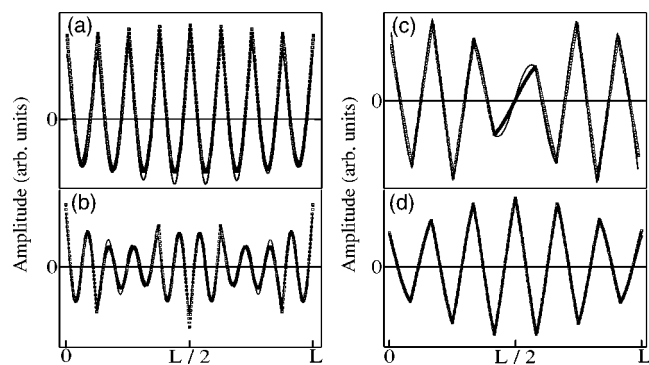


FIG. 10. The amplitudes of the Timoshenko Beam Theory (continuous line) and the Bernoulli-Euler approach (squares) are very similar. (a), (b), (c) and (d) correspond to the same cases of Fig. 9.

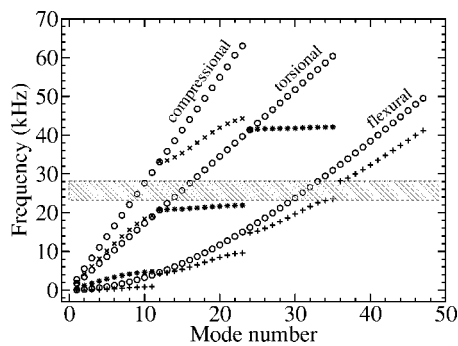


FIG. 11. Measured band spectra for compressional (crosses), torsional (stars) and flexural vibrations (pluses) of a locally periodic aluminum rod of length $L=0.9105$ m and radius 6.35 mm and $N=12$ unit cells. The notches have radius $r=1.3$ mm and width $\epsilon=0.3$ mm. The circles correspond to experimental values of a uniform aluminum rod of the same length and radius. The frequencies in the shaded band form a common stop band for the three types of vibration.

time. This was possible thanks to a noncontact electromagnetic acoustic transducer introduced recently.¹³ It was shown that a banded spectrum arises as the number of unit cells grows. This turns out to be the case also from the theoretical point of view. Both Timoshenko theory and the simpler Bernoulli-Euler approach predict also a banded spectrum for the normal-mode frequencies. The fit to the experimental values is, however, much better if the first of these theories is used.

As mentioned in the Introduction, bands also emerge for locally periodic rods when dealing with compressional and torsional vibrations. The experimental results for these cases⁷ are compared to the values obtained in this work for flexural vibrations for the particular case $N=12$ in Fig. 11. It is observed that the gap is produced in all types of vibrations when the Bragg condition holds. In this figure, it is also clear that a common gap (shaded region) in the frequency spectrum appears, so a very efficient isolation vibration is possible for these frequency values.

Compressional and torsional vibrations are described by second-order differential equations whereas flexural vibrations require a fourth-order differential equation, so one could conclude that bands appear due to the local invariance with respect to translations that the system shows and not to the peculiarities of the equations governing them.

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